

ENR 4 RADIO NAVIGATION AIDS/SYSTEMS

ENR 4.1 RADIO NAVIGATION AIDS - EN-ROUTE

Name of station	ID	Frequency (CH)	Hours of operation	Coordinates	Elevation of DME Transmitting antenna	Remarks
1	2	3	4	5	6	7
Basco NDB	BS	276 KHZ	2300 - 0700	202705.3N 1215829.3E	Nil	Output: 50 watts. 320 M left of RWY 06 CL.
Bicol DVOR/DME	BIK	113.6MHZ (CH83X)	H24	130651.0N 1234035.1E	100 FT	DVOR output PWR: 100 watts. DME output PWR: 1000 watts. Usable DIST: 90 NM. DIST FM CL: 202 M. DIST FM THR 23: 655 M. Annual rate of change: 6' increasing. Restriction: DVOR unusable FM Radial 005 to Radial 010.
Butuan DVOR/DME	BN	112.5MHZ (CH72X)	H24	085707.5N 1252901.6E	Nil	300 M left of CL. VOR output - 100 watts. DME output - 1 KW.
Cabanatuan DVOR/DME	CAB	112.7MHZ (CH74X)	H24	152856.2N 1210127.5E	53.9359 M / 176.964 FT	Coverage: DVOR/DME - 100 NM. VOR unusable on: Radial 350 to Radial 070 MRA 8500 FT 40 NM, Radial 040 to Radial 100 MRA 9500 FT 40 NM, Radial 100 to Radial 140 MRA 8500 FT 40 NM, Radial 140 to Radial 170 MRA 6000 FT 40 NM, Radial 170 to Radial 220 MRA 3000 FT 40 NM, Radial 200 to Radial 290 MRA 6000 FT 40 NM, Radial 290 to Radial 350 MRA 4000 FT 40 NM.
Cauayan DVOR/DME	CUY	116.5MHZ (CH112X)	0000 - 0800	165553.8N 1214453.1E	Nil	Coverage: DVOR - 90 NM. DME - 70 NM. 376 M FM start of RWY 12 and 150.85 M right side of RWY 12 CL.

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Clark DVOR/DME	CIA	113.1MHZ (CH78X)	H24	151112.8N 1203338.3E	500 FT	Managed by: Bases Conversion and Development Authority (BCDA). Maintained by: CAA of the Philippines. DVOR - 100 watts. DME - 1 KW.
Cotabato VOR/DME	COT	113.7MHZ (CH84X)	H24	071003.1N 1241232.8E	55.712 M / 182.781 FT	VOR - 100 watts. DME - 1 KW. VOR unusable on Radial 120 to Radial 170 beyond 30 NM below 8500 FT; Radial 170 to Radial 280 beyond 30 NM below 12000 FT.
Davao DVOR/DME	DAO	112.1MHZ (CH58X)	H24	070752.5N 1253914.3E	100 FT	409 M FM RWY 23 and 105.5 M left of RWY 23 CL. VOR - 200 watts. DME - 1 KW. Unusable on Radial 128 beyond 35 NM below 4500 FT.
Iloilo DVOR/DME	IOO	116.3MHZ (CH110X)	H24	105007.4N 1222931.5E	53.279 M / 174.8 FT	Coverage: DVOR - >100 NM. DME - >78 NM. 1445.95 M FM THR of RWY 02, 1004.05 M FM THR of RWY 20 and 222 M FM RWY CL; VOR - 100 watts. DME - 1 KW.
Jomalig DVOR/DME	JOM	116.7MHZ (CH114X)	0900 - 1700	144217.5N 1221947.9E	23.3859 M / 76.7291 FT	570 M FM start of RWY 23 and 130 M right of RWY 23 CL. Coverage: DVOR - 90 NM. DME - 70 NM.
Kalibo DVOR/DME	KLO	113.0 MHZ (CH77X)	H24	114100.8N 1222259.8E	100 FT	Coverage: DVOR/DME - 100 NM. 298 M FM start of RWY 23 and 220 M left of RWY 23 CL.
Laguindingan DVOR/DME	LGD	116.1 MHZ (CH108X)	H24	083648.9N 1242708.2E	200 FT	DME Output PWR: 1000 watts. DVOR Output PWR: 100 watts

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Laoag DVOR/DME	LAO	112.9MHZ (CH76X)	H24	181043.7N 1203144.6E	100 FT	Coverage: DVOR/DME - 90 NM. 1117 M from start of RWY 19 and 242.05 M right of RWY 19 center line.
Mactan DVOR/DME	MCT	114.3MHZ (CH90X)	H24	101848.8N 1235917.9E	100 FT	500 M from THR RWY 22L and 220 M right of center line. VOR - 100 watts. DME - 1 KW. VOR unusable Radial 300 to Radial 320 beyond 40 NM below 8500 FT.
Manila DVOR/DME	MIA	114.4MHZ (CH91X)	H24	143028.7N 1210118.2E	100 FT	193.66 M from RWY 06/24 center line and 117.83 M from extended RWY 13/31 center line. DVOR output power: 100 watts. DME output power: 1000 watts. Restriction: VOR Radial 090 to Radial 100 and Radial 114 to Radial 120 unusable beyond 50 NM.
Masbate DME	MBT	CH104X	2000 - 1130	122209.2N 1233749.8E	54.631 FT	DME reception limited to FL060 at 20 NM for Radial 165 to Radial 185. Coverage: 150 NM.
Naga DVOR	NGA	114.7 MHZ	2200 - 1000	133456.0N 1231610.3E	Nil	415 M from start of RWY 04 and 130 M right of RWY center line. Unusable on Radial 040 and Radial 070 beyond 20 NM below FL170.
Naga DME	NGA	CH94X	2200 - 1000	133455.5N 1231610.0E	200 FT	Coverage: DVOR - 100 NM. DME - 60 NM.

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Puerto Princesa DVOR/DME	PR	115.9MHZ (CH106X)	H24	094423.3N 1184613.5E	100 FT	DVOR Output Power: 100 watts. Usable distance: 40 NM. Distance from center line: 204 M / 668 FT. Distance from THR 27: 204 M / 669 FT. Caution: Unusable on Radial 230 to Radial 240 beyond 30 NM below 8500 FT, Radial 240 to Radial 260 beyond 40 NM below 7500 FT, Radial 260 to Radial 270 beyond 33 NM below 8500 FT, Radial 270 to Radial 290 beyond 35 NM below 1100 FT, Radial 290 to Radial 300 beyond 35 NM below 9500 FT, Radial 300 to Radial 310 beyond 40 NM below 8500 FT, Radial 310 to Radial 323 beyond 40 NM below 6500 FT, Radial 323 to Radial 345 beyond 40 NM below 6000 FT, Radial 345 to Radial 018 beyond 34 NM below 5500 FT.
Roxas DVOR/DME	ROX	112.9MHZ (CH76X)	2200 - 1100	113548.9N 1224518.4E	44.314 FT	Coverage: DVOR/DME - 100 NM.
San Jose DVOR/DME	SAJ	117.1 MHZ (CH118X)	H24	122137.1N 1210235.1E	Nil	DVOR - 100 watts. DME - 1 KW. 224 M from North end RWY and 72 M from RWY center line.
Subic Bay DVOR/DME	SBA	113.5MHZ (CH82X)	H24	144610.6N 1201330.3E	200 FT	Located at Grande Island. Unusable from Radial 275 to Radial 295 due to obstruction.
Tacloban DVOR/DME	TAC	115.5MHZ (CH102X)	H24	111355.2N 1250133.6E	100 FT	Coverage: VOR - 60 NM. DME - 40 NM.

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Tambler DVOR/DME	GSA	114.5MHZ (CH92X)	H24	060432.4N 1250531.9E	600 FT	DME - 1 KW. DVOR - 100 watts. DVOR unusable on Radial 245 to Radial 306 and Radial 353 to Radial 360 beyond 15 NM due obstruction.
Zamboanga DVOR/DME	ZAM	113.9MHZ (CH86X)	H24	065528.9N 1220319.7E	100 FT	Coverage: DVOR/DME - 50 NM. DVOR/DME unusable on Radial 270 to Radial 360 and Radial 000 to Radial 015 beyond 40 NM FM ZAM and below FL100. Radial 315 to Radial 360 and Radial 000 to Radial 015 between 25 NM to 40 NM FM ZAM and below FL100. Radial 320 to Radial 355 within 12 NM FM ZAM and below FL060.

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